

# SOC Investigation Report - LSASS Memory Access Detected

*No confirmed malicious Indicators of Compromise (IOCs) were identified during this investigation.*

|                       |                           |
|-----------------------|---------------------------|
| <b>Client</b>         | Kerning City Dental (KCD) |
| <b>Prepared by</b>    | Graham McKean             |
| <b>Date</b>           | August 2026               |
| <b>Classification</b> | CONFIDENTIAL              |
| <b>Severity</b>       | Medium                    |

## Findings

|                         |                                               |
|-------------------------|-----------------------------------------------|
| <b>Alert</b>            | KCD - Endpoint - LSASS Memory Access Detected |
| <b>Classification</b>   | Inconclusive - Likely Benign                  |
| <b>Severity</b>         | Medium                                        |
| <b>Detection Source</b> | Splunk SIEM                                   |
| <b>MITRE ATT&amp;CK</b> | T1003 - Credential Dumping (Sysmon Rule Tag)  |
| <b>Affected Host</b>    | DESKTOP-RHD61K.kerningcitydental.ca           |
| <b>Source Process</b>   | C:\Windows\Sysmon.exe                         |
| <b>Target Process</b>   | C:\Windows\System32\lsass.exe                 |
| <b>Granted Access</b>   | 0x1fffff                                      |
| <b>Source User</b>      | NT AUTHORITY\SYSTEM                           |

## Investigation Summary

The investigation validated the original Sysmon Event ID 10 (Process Access) that generated the Splunk alert. The event confirmed that C:\Windows\Sysmon.exe, running under the NT AUTHORITY\SYSTEM security context, accessed C:\Windows\System32\lsass.exe with a GrantedAccess value of 0x1fffff.

Review of the available surrounding telemetry did not identify corroborating evidence of credential dumping. Searches for common credential-dumping indicators and utilities, including Mimikatz, ProcDump, comsvcs.dll, MiniDumpWriteDump, sekurlsa and related activity, did not identify supporting evidence of credential theft or LSASS memory-dump creation.

The available telemetry did not allow the digital signature, file hash, version, publisher or deployed configuration of Sysmon.exe to be independently validated. Additional endpoint process information was sought, but no corresponding process-creation data was returned. The legitimacy of the Sysmon binary itself therefore could not be conclusively established.

No corroborating evidence of endpoint compromise was identified. However, because the source binary could not be fully validated and the available evidence does not establish whether comparable LSASS-access events recurred outside the reviewed event context, the alert is assessed as Inconclusive - Likely Benign rather than conclusively Benign.

### Timeline

| Time (UTC)               | Event                                                                                                                         |
|--------------------------|-------------------------------------------------------------------------------------------------------------------------------|
| 03 Aug 2026 03:13:40     | Sysmon Event ID 10: Sysmon.exe accessed lsass.exe with GrantedAccess 0x1fffff.                                                |
| Investigation            | No supporting evidence of credential dumping or LSASS memory-dump creation identified.                                        |
| Investigation limitation | Recurrence of similar LSASS-access events outside the reviewed event context was not established from the available evidence. |

### Investigation

Validated the original Sysmon event, confirmed the source and target processes, reviewed surrounding telemetry for credential-dumping indicators, and assessed the available endpoint evidence. Nearby Windows servicing activity was not used as a causal explanation because the available evidence did not demonstrate a direct relationship to the Sysmon-to-LSASS access.

### 5Ws and 1H

|       |                                                                                                                                                               |
|-------|---------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Who   | NT AUTHORITY\SYSTEM.                                                                                                                                          |
| What  | Sysmon Process Access event against lsass.exe.                                                                                                                |
| When  | 03 Aug 2026 03:13:40 UTC. Whether similar access recurred outside the reviewed event context was not established.                                             |
| Where | DESKTOP-RHD61K.kerningcitydental.ca.                                                                                                                          |
| Why   | The event matched a Sysmon credential-dumping detection rule. No malicious intent or confirmed legitimate cause was established from the available telemetry. |
| How   | Sysmon.exe opened a handle to lsass.exe with GrantedAccess 0x1fffff.                                                                                          |

### Observed Indicators

|                |                                     |
|----------------|-------------------------------------|
| Host           | DESKTOP-RHD61K.kerningcitydental.ca |
| Source Process | C:\Windows\Sysmon.exe               |

|                       |                               |
|-----------------------|-------------------------------|
| <b>Target Process</b> | C:\Windows\System32\lsass.exe |
| <b>GrantedAccess</b>  | 0x1fffff                      |
| <b>Event ID</b>       | 10                            |

## Confirmed Malicious IOCs

None identified.

## Assessment

Alert Classification: Inconclusive - Likely Benign.

The investigation confirmed a genuine Sysmon Event ID 10 in which C:\Windows\Sysmon.exe, running as NT AUTHORITY\SYSTEM, accessed C:\Windows\System32\lsass.exe with GrantedAccess=0x1fffff. The T1003 tag identifies the detection logic and does not by itself prove credential dumping. No corroborating credential-dumping or LSASS memory-dump activity was identified. However, the Sysmon binary's signature, hash, version, publisher and deployed configuration could not be independently validated. The available evidence therefore supports a likely benign interpretation, but not a conclusive benign closure.

## Recommendations

- Retain the Sysmon Event ID 10 and associated Splunk telemetry as investigation evidence.
- Validate the digital signature, file hash, version and publisher of C:\Windows\Sysmon.exe if endpoint access or suitable file telemetry becomes available.
- Review the deployed Sysmon configuration to determine whether the observed LSASS access is expected under the configured monitoring policy.
- Review a wider time window for repeated Sysmon-to-LSASS Event ID 10 activity and document whether the event is isolated or recurring.
- Continue monitoring for LSASS access involving unexpected or non-system processes and investigate any associated credential-dumping indicators.

## Investigation Limitations

The available telemetry did not allow the digital signature, file hash, version, publisher or deployed configuration of Sysmon.exe to be independently validated. No corresponding process-creation data for Sysmon.exe was returned, so its parent process, command line and additional file metadata could not be independently confirmed. The evidence reviewed also did not establish whether similar LSASS-access events recurred outside the reviewed event context. Findings and conclusions are limited to the telemetry available during the investigation.